

2013-2014 – Master 2 – Macro I

## Lecture 1 : Optimal Control

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(based on Gilles Saint-Paul lecture notes)

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# Disclaimer

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## 0. Introduction

- ▶ What is optimal control? It is an optimization technique aimed at computing the optimal trajectory of a dynamical system.
- ▶ At each date this system is in a certain position described by a vector of **state variables**. These state variables cannot be determined directly by us because they can only change gradually.
- ▶ We can change the trajectory because the evolution of the system is affected by the **control variables** that we can freely choose at a point in time.
- ▶ This class of problems is called an optimal control problem and we can solve it by using an extension of the Kuhn-Tucker theorem called the maximum principle.

# 1. Discrete time

## The problem

- ▶ Consider the following maximization problem :

$$\begin{aligned} \text{Max } V &= \sum_{i=0}^N U_i(K_i, C_i) \\ \text{s.t. } K_0 &= \bar{K}_0 \\ K_{i+1} &= G(K_i, C_i), i = 0, \dots, N \\ K_{N+1} &\geq 0 \end{aligned}$$

- ▶  $N + 1$  dates indexed by  $i = 0, \dots, N$ .
- ▶ Vector of state variables given by  $K = (k_1, \dots, k_p)^\top \in \mathbb{R}^p$ .
- ▶ Vector of control variables  $C = (c_1, \dots, c_q)^\top \in \mathbb{R}^q$ .
- ▶  $G()$  determines the evolution of the system.
- ▶ We also have an initial and a final condition

# 1. Discrete time

Khun & Tucker

- ▶ Call  $\lambda_i = (\lambda_{1i}, \dots, \lambda_{pi})^\top$  the vector of Lagrange multipliers associated with the constraint  $K_{i+1} = G(K_i, C_i)$  and  $\bar{\lambda}$  the vector of Lagrange multipliers associated with the constraint  $K_{N+1} \geq 0$ .
- ▶ Then  $\lambda_{ji}$  can be interpreted as the marginal value of one extra unit of  $k_j$  at date  $i$ , i.e. the effect on  $V$  *at the optimum* if  $k_{ji}$  increases exogenously by one small unit.
- ▶ Denote by  $\nabla_X F()$  the gradient of function  $F$  with respect to vector  $X$ .
- ▶ Treat  $K$ ,  $C$ , and  $G()$  as column vectors, while the gradient is defined as a line vector, and can be extended to each component of a vector-valued function.
- ▶ For example,  $\nabla_C G = (\frac{\partial G}{\partial c_1}, \dots, \frac{\partial G}{\partial c_q})$  is a  $p \times q$  matrix.
- ▶ A  $\top$  denotes the transpose and products are matrix products. Thus the scalar product of two column vectors  $X$  and  $Y$  is  $X^\top Y$ . A dot  $(\cdot)$  denotes element-by-element multiplication.

# 1. Discrete time

## Lagrangian

- ▶ The Lagrangian is equal to

$$\mathcal{L} = \sum_{i=0}^N U_i(K_i, C_i) + \sum_{i=0}^N \lambda_i^\top (G(K_i, C_i) - K_{i+1}) + \bar{\lambda}^\top K_{N+1}.$$

- ▶ First-order conditions (FOC) :

$$\nabla_{C_i} \mathcal{L} = 0 \iff \nabla_{C_i} U_i + \lambda_i^\top \nabla_{C_i} G = 0, i = 0, \dots, N$$

$$\nabla_{K_i} \mathcal{L} = 0 \iff \nabla_{K_i} U_i - \lambda_{i-1}^\top + \lambda_i^\top \nabla_{K_i} G = 0, i = 1, \dots, N$$

$$\nabla_{K_N} \mathcal{L} = 0 \iff \lambda_N = \bar{\lambda}.$$

- ▶ **Exercise 1** : Check the conformity of the above matrix products. How many equations are defined in each of these conditions ?
- ▶ Finally we also have the complementary slackness condition for the inequality constraint  $K_{N+1} \geq 0$  :

$$\bar{\lambda} \cdot K_{N+1} = 0, \text{ or equivalently } \lambda_N \cdot K_{N+1} = 0.$$

## 2. Continuous time

### The problem

- ▶ We can now derive the optimality conditions of a similar problem in continuous time by approximating it by a discrete time problem and using the preceding first-order conditions.
- ▶ The continuous-time problem is

$$\begin{aligned} \text{Max } V &= \int_0^T U(K_t, C_t, t) dt \\ \text{s.t. } K_0 &= \bar{K}_0 \\ \dot{K}_t &= F(K_t, C_t) \\ K_T &\geq 0. \end{aligned}$$

## 2. Continuous time

### Discrete time approximation

- ▶ We split the  $[0, T]$  interval into  $N$  intervals of equal length  $\Delta t = \frac{T}{N}$  and approximate it by the following discrete problem :

$$\text{Max} \sum_{i=0}^N U_i(\tilde{K}_i, \tilde{C}_i) \Delta t,$$

subject to

$$\begin{aligned}\tilde{K}_{i+1} &= \tilde{K}_i + F(\tilde{K}_i, \tilde{C}_i) \Delta t; \\ \tilde{K}_0 &= \bar{K}_0; \\ \tilde{K}_{N+1} &\geq 0.\end{aligned}$$

- ▶ the tildas refer to  $K$  and  $C$  expressed as a function of then number of iterations rather than time. Thus  $\tilde{K}_i = K_{i\Delta t}$  and similarly for  $C$ .
- ▶ By definition  $U_i(\tilde{K}_i, \tilde{C}_i) = U(K_{i\Delta t}, C_{i\Delta t}, i\Delta t)$ .

## 2. Continuous time

$$N \rightarrow \infty$$

- ▶ As  $N \rightarrow \infty$ ,  $\sum_{i=0}^N U_i(\tilde{K}_i, \tilde{C}_i)\Delta t \rightarrow \int_0^T U(K_t, C_t, t)dt$  and the evolution equation  $\tilde{K}_{i+1} = \tilde{K}_i + F(\tilde{K}_i, \tilde{C}_i)\Delta t$  converges to  $\dot{K}_t = F(K_t, C_t)$ .
- ▶ Thus the discrete problem approximates the continuous one as  $N \rightarrow \infty$ .

## 2. Continuous time

### FOC

- ▶ The previous section gives us the FOC for the discrete problem.
- ▶ Note that the  $G()$  function is now given by  $G(\tilde{K}_i, \tilde{C}_i) = \tilde{K}_i + F(\tilde{K}_i, \tilde{C}_i)\Delta t$ .
- ▶ Put a tilda on the Lagrange multipliers whenever expressed as a function of  $i$  rather than  $t$  and get :

$$\begin{aligned}\Delta t.(\nabla_{\tilde{C}_i} U_i + \tilde{\lambda}_i^\top \nabla_{\tilde{C}_i} F) &= 0, i = 0, \dots, N \\ \nabla_{\tilde{K}_i} U.\Delta t - \tilde{\lambda}_{i-1}^\top + \tilde{\lambda}_i^\top (1 + \nabla_{\tilde{K}_i} F.\Delta t) &= 0, i = 1, \dots, N \\ \tilde{\lambda}_N \cdot \tilde{K}_{N+1} &= 0.\end{aligned}$$

## 2. Continuous time

### FOC (continued)

- ▶ Take  $N \rightarrow \infty$ , implying  $\Delta t \rightarrow 0$ ;
- ▶ Ignore second order terms in  $\Delta t$  and reexpress variables as a function of  $t$  rather than  $i$ ,
- ▶ We get the following, which must hold at all dates  $t$  :

$$\begin{aligned}\nabla_C U + \lambda^\top \nabla_C F &= 0 \\ \nabla_K U + \lambda^\top \nabla_K F &= -\dot{\lambda}^\top \\ \lambda_T \cdot K_T &= 0.\end{aligned}\tag{1}$$

## 2. Continuous time

### Hamiltonian

- ▶ Define

$$H_t = U(K_t, C_t, t) + \lambda_t^\top F(K_t, C_t).$$

- ▶  $H_t$  is called the **Hamiltonian**. We can express it as a function  $H_t = H(K_t, C_t, \lambda_t, t)$ .
- ▶ Then the preceding optimality conditions can be rewritten

$$\begin{aligned}\nabla_C H &= 0 \\ \nabla_K H &= -\dot{\lambda}^\top \\ \lambda_T \cdot K_T &= 0.\end{aligned}$$

- ▶ The first two conditions are called the first-order conditions. The last one is called the *transversality condition*. These conditions are necessary for optimality.

## 2. Continuous time

### Notes

- ▶ **Note** : The transversality condition is explicitly associated with the terminal inequality constraint  $K_T \geq 0$ . Other terminal conditions will deliver different transversality conditions.
- ▶ **Note** : If  $K$  and  $C$  are scalars, so is  $\lambda$ , and the FOCs are simply rewritten as  $H'_C = 0$ ,  $H'_K = -\dot{\lambda}$ ,  $\lambda_T K_T = 0$ .
- ▶ **Exercise 2** : Suppose there are two state variables,  $k_t$  and  $s_t$ , and two control variables,  $c_t$  and  $z_t$ . Write down the first-order conditions in scalar form, rather than vector form. How many conditions do we get?
- ▶ **Exercise 3** : Show that if there is no terminal condition then the transversality condition is  $\lambda_T = 0$ .
- ▶ **Exercise 4** : What is the transversality condition if the terminal condition is an equality constraint  $K_T = \bar{K}$ ?

## 2. Continuous time

### Remarks

- ▶ **Remark 1 :** We can also write  $(\nabla_{\lambda} H)^{\top} = F(K_t, C_t) = \dot{K}$ .
- ▶ **Remark 2 :** The Lagrange multiplier  $\lambda_t$  is also called the co-state variable.
- ▶ **Remark 3 :** Assume  $U(.,.)$  is time-invariant. Then along the optimal trajectory, we have

$$\dot{H} = \nabla_C H \cdot \dot{C} + \nabla_K H \cdot \dot{K} + \nabla_{\lambda} H \cdot \dot{\lambda} = 0 \times \dot{C} - \dot{\lambda}^{\top} \dot{K} + \dot{K}^{\top} \dot{\lambda} = 0.$$

The value of the Hamiltonian is invariant along the optimal trajectory.

## 2. Continuous time

### Remarks (continued)

- ▶ **Remark 4 :** The actual necessary conditions for optimality are a bit different from above : instead of (1), we need that the Hamiltonian is maximized with respect to  $C$  at all dates. If the Hamiltonian is concave, the two are equivalent. If the Hamiltonian is smooth, (1) is still necessary, but weaker than actual maximization of  $H$ . Finally there are cases where the solution is "bang-bang" because the problem is convex rather than concave, in which case  $C$  will be at a corner solution. In such a case (1) is not necessary, but it can be restored in some form by explicitly adding the constraints for the interval(s) of variation of  $C$ .
- ▶ **Remark 5 :** To the best of my knowledge, I have not found any result saying that these conditions are sufficient (we have not derived the SOC's). We just know that the optimal trajectory must satisfy them. In most of our applications this determines a unique trajectory so we are confident it is the optimal one.

### 3. Discounting

#### FOC

- ▶ In most economic applications  $U$  is not time-invariant because of discounting :  $U(K_t, C_t, t) = e^{-\rho t} u(K_t, C_t)$ .
- ▶ Optimality conditions are :

$$\begin{aligned}\nabla_C U + \lambda^\top \nabla_C F &= 0 \\ \nabla_K U + \lambda^\top \nabla_K F &= -\dot{\lambda}^\top \\ \lambda_T \cdot K_T &= 0.\end{aligned}$$

- ▶ Using the expression of  $U$  :

$$\begin{aligned}e^{-\rho t} \nabla_C u + \lambda_t^\top \nabla_C F &= 0 \\ e^{-\rho t} \nabla_K u + \lambda_t^\top \nabla_K G &= -\dot{\lambda}_t \\ \lambda_T \cdot K_T &= 0.\end{aligned}$$

### 3. Discounting

#### FOC (continued)

- ▶ Let's introduce a current (and not discounted) co-state variable  $\mu$  :

$$\lambda_t = e^{-\rho t} \mu_t.$$

- ▶ Then FOC then be rewritten :

$$\begin{aligned}\nabla_C u + \mu_t^\top \nabla_C F &= 0 \\ \nabla_K u + \mu_t^\top \nabla_K G &= -\dot{\lambda}_t = \rho \mu_t - \dot{\mu}_t \\ e^{-\rho T} \cdot \mu_T \cdot K_T &= 0.\end{aligned}$$

## 4. Infinite Horizon

- ▶ the approach can be applied to infinite horizon models by letting  $T$  go to infinity.
- ▶ The only change is that we now take the limit in the transversality condition :

$$\lim_{T \rightarrow \infty} e^{-\rho T} \mu_T \cdot K_T = 0.$$

- ▶ **Remark :** We will get a different result if we were simplifying the transversality condition first, by getting rid of  $e^{-\rho T} \neq 0$ , and then taking the limit. With a finite horizon, it means that both the contemporaneous ( $\mu_T$ ) and discounted at  $t = 0$  ( $e^{-\rho T} \mu_T$ ) values of  $K$  must vanish. But in infinite horizon, the contemporaneous value of  $K$  may remain positive asymptotically, while its value at  $t = 0$  is equal to zero due to discounting. Since we are optimizing at  $t = 0$ , it is the latter that matters for the transversality condition.